

Attracting talent by building it in house.

Attracting and retaining qualified candidates in today's competitive climate is becoming more and more difficult. One possible solution? Build a foundation to attract, support, and grow Junior Developers in house. The Web & Application Services team is leveraging the ITS Summer Internship Program to jump start this process.

There is a global talent shortage and rising wages

The demand for experienced programmers is far outpacing the supply, and the outlook is getting worse. According to [U.S. Labor statistics](#) as of December 2020, the global talent shortage amounted to 40 million skilled developers and by 2030, the global talent shortage is expected to reach 85.2 million.

Additionally, the pandemic has greatly impacted the very nature of how we approach our work. For example, remote work has become not just an option, but is now an expectation. For developers, remote work has become the norm.

In an environment where developers are in such high demand, and can chose to work for *anyone*...from *anywhere* in the world, LSA can no longer compete on the basis of salary and benefits. The job market for more experienced (mid-level and senior) software developers is even more competitive and favorable for applicants.

The opportunity

Dave Chmura from LSA Technology Services W&ADS Team saw an opportunity. He set out to build a talent pipeline through hiring and training interns.

"I see the enormous upside in working with newer, less experienced developers," he explained. "If you are able to get buy-in and support to really focus and invest in your juniors, it can pay huge dividends and have benefits that ripple throughout the whole team."

Instead of adding interns to supplement the team, Chmura proposed developing an entire program, focused 100% on training them. He conceived of a program that would function as a kind of a bootcamp, training novices how to develop full scale web applications using Ruby on Rails, the framework that a team within W&ADS develops in.

"It isn't only about adding talent. By framing our mission on the instruction, we were also building in an automatic review of how we approach building software. Juniors don't

have the same assumptions or history that we carry. When we teach them how we do things we are naturally encouraged to re-examine the 'why' behind it. It both gives us an opportunity to explore if there's a better way plus, just the act of explaining it deepens our understanding." Chmura continued.

Pat Belden, director of the W&ADS team saw the potential right away. By bolstering our ability to recruit and train more junior developers, W&ADS could hedge against the talent shortage, while simultaneously improving how the team functions.

ITS Internship Program

For the first time this year, LSA has partnered with Information and Technology Services (ITS) at the University of Michigan by participating in their annual summer internship program. ITS Internships are paid, full-time positions that provide an opportunity for students to gain valuable experience while making connections in the professional field they are considering for a career. Interns have the opportunity to work on meaningful projects in a structured and supervised learning program. The program runs for 14 weeks, starting in May and ending the first week in August.

Dave Chmura led the charge for the W&ADS Team and was very intentional in how he wrote the descriptions for our internship positions, focusing on potential and enthusiasm instead of solely on acquired skills. (** ***I can make the job description available online if you want to link to it.***)

"The qualities we were looking for had nothing to do with skill sets...well, very little anyway. We were looking for the candidates who had the best opportunity to increase their learning the most! We were also mindful that we were assembling a team. It wasn't just the individual's qualifications, but how well they could learn, and learn together."

Not just a summer fling...

From the beginning, the plan was to create a dynamic and enduring program. The program is set up so that it not only teaches the interns how to do iterative design and development, it benefits from the same principles we are teaching.

"I wanted to develop a virtuous cycle where we could bring two early career students onto our team, teach them to program in the framework that our team uses, hire them to work with us as student employees during the following school year, and hopefully invite them back the following summer to help us train the next batch of interns."

"By structuring this cycle, it affords our interns the opportunity for a broader learning experience. They are not just learning to code and develop applications; they are learning to communicate effectively, to develop and structure lessons. We don't just want to train workers, we want to coach leaders."

After screening over 100 candidates and conducting 16 interviews, Dave was able to offer positions to Henry Teng and Debbie Shih on the Web & Application Development Services team.

"Most employers focus on finding candidates with specific skill sets to accomplish tasks. I wanted to turn that idea on its head." Chmura explained. "Rather than trying to find a candidate that checked off some 'qualifications', we focussed on finding candidates who were thirsty and eager to learn."

Dave also volunteered to participate and contribute to the ITS orientation planning committee so that he could quickly learn how the program was structured, and to tailor our approach to take the most advantage possible.

Henry and Debbie are working on a Rails application that will provide the W&ADS team a "Dashboard" to track LSA's applications. So far the experience has been FANTASTIC for both the W&ADS Team and our interns. How do Henry and Debbie like working with the W&ADS Team? See some of the comments below!

In previous projects, my process was always very unstructured and often disorganized. I wanted to observe a real developer's process and way of thinking so that I could integrate it into my procedure.

I like being able to work on meaningful software projects for our school community, since I have really liked my time at UMich and would like to give back in my own way.

The fact that I'm actually passionate about the stuff I'm learning about definitely makes it ten times more fun. I often lose track of time when I'm working on my web application because I get so into it, which made me quite surprised. I'm grateful that I have a supervisor that puts up with my constant questions and team members that are always willing to help out.

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